

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Contributed to the preparation of feasibility and detailed design reports for water infrastructure projects, including drafting **drawings** and performing initial **calculations**.
- Assisted senior engineers in preparing documentation for statutory approvals, such as **planning permissions** and wayleaves, ensuring compliance with regulatory requirements.
- Supported the development of detailed project plans by contributing to the allocation of **resources** and outlining preliminary **schedules** for water engineering initiatives.
- Collaborated with cross-functional teams to gather requirements for water infrastructure projects, facilitating effective communication between project managers and **clients**.
- Contributed to site-based activities and conducted site visits across Ireland to gain practical exposure to water engineering projects, documenting observations in **Confluence**.
- Participated in Agile/Scrum ceremonies and utilized **Jira** for defect tracking and task management, ensuring efficient workflow within the engineering team.

SKILLS

Engineering Principles – Water Engineering, Civil Engineering, Environmental Engineering, Infrastructure Projects, Feasibility Studies, Design Reports, Calculations, Specifications, Contract Documents, Construction Stage Contract Administration, Project Plans, Budgets, Schedules, Resource Allocation, Project Risk Mitigation, Health & Safety Legislation, Quality Management Systems.

Software & Tools – AutoCAD, Civil 3D, GIS, ArcGIS, QGIS, Epanet, Synergi, MicroDrainage, HEC-RAS, Python, Jira, Confluence, Git, Jenkins.

Data & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling, SQL, R, RStudio, pandas, NumPy, scikit-learn, Statistical Analysis, Exploratory Data Analysis, Business Intelligence.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD, Jenkins.

Programming Languages – Java, Python, SQL, R.

Databases – PostgreSQL, MySQL, Oracle.

PROJECTS

Water Network Analysis and Design Tool (HCL Technologies)

- Developed Python scripts to automate data ingestion and initial analysis of water network parameters from various input formats, reducing manual processing time by approximately **30%**.
- Designed and implemented a basic data model using **PostgreSQL** to store network topology and flow data, facilitating efficient querying for hydraulic simulations.
- Created preliminary visualizations of network flow data using **Power BI** to support design report generation and communicate key findings to internal project teams.

Environmental Data Reporting Dashboard (MSc Project)

- Built an interactive dashboard in **Power BI** to visualize environmental impact data for a simulated infrastructure project, incorporating **DAX measures** for key performance indicators.
- Performed ETL processes using **Power Query** to clean and transform disparate data sources into a unified dataset suitable for analysis and reporting.
- Presented the final dashboard to academic stakeholders, demonstrating the ability to translate complex data into actionable insights for project decision-making.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)